NumNum: An Asian-Focused Restaurant Discovery Platform

NYC-Based · Community-Driven · Asian Cuisine Focused

PROJECT OVERVIEW

What is NumNum?

NumNum is a web application designed to help food lovers in NYC and nearby NJ efficiently discover, compare, and share Asian restaurants — all in one structured, community-powered platform.

INTENDED USERS

- *International students in NYC who frequently search for Asian food*
- *Young professionals in NYC/NJ looking for efficient dining decisions*
- *Visitors exploring Asian food scenes in NYC/NJ*
- *Community members wanting to share dining experiences*

CORE OBJECTIVES

- Structured multi-dimensional search & filter by cuisine sub-type, area, price, rating, vibe, and signature dish
- Dish-level detail pages with community ratings, photos, and menu highlights — not just restaurant-level scores
- Community social feed where users share dining posts, ask for recommendations, and build trusted food circles
- Seamless NYC ↔ NJ switching with map view and location-aware restaurant discovery

COMPETITIVE LANDSCAPE

Why This Platform?

Platform	Cuisine Depth	Community	Dish Detail
Google Maps	Low	None	Medium
Yelp	Low	Low	Medium
Rednote	High	High	Low
Our Platform	**High**	**High**	**High**

Gap in the market: Existing platforms are broad but generic, or social but unstructured. NumNum combines Asian cuisine depth, community trust, and dish-level decision support.

Key Features and Functionalities (1/2)

Smart Discovery & Social Recommendation

Feature 1: Smart Multi-Dimensional Filter & Structured Restaurant Cards

FUNCTIONALITY

Helps users efficiently discover Asian restaurants across NYC & nearby NJ through structured search, filtering, and comparison — instead of scattered posts or generic listings.

INFO PROVIDED

Name · Cuisine Sub-type · Location · Price Range · Community Rating · Dining Vibe Tags · Walk-in/Reservation Info · Top Dish / Signature Dish

INTERACTIVE

- Filter by location, cuisine, price, minimum rating, vibe, and practical dining options
- Apply soft tags (photogenic, hidden gem, walk-in friendly, local favorite, late night)
- Sort results dynamically · Switch NYC ↔ NJ regions
- Click restaurant cards → detail pages · Toggle list/map view

USER WORKFLOW

Enter search / select filters → platform updates results instantly → compare structured cards (cuisine, price, distance, rating, top dish) → click into restaurant detail page.

Why it matters: Reduces decision fatigue and makes restaurant discovery faster, clearer, and more structured.

Feature 2: Community Social Space — Posts, Discussions & Friend Recs

FUNCTIONALITY

Adds a social recommendation layer called Feed, where users share dining experiences, ask for suggestions, and discover restaurants through trusted friends or community — not anonymous ratings alone.

INFO PROVIDED

User posts & dining experiences · Dish-specific comments · Friend/community trust signals · Personal or shared lists · Community discussions & recommendations

INTERACTIVE

- Create posts with dining notes & food photos
- Message friends asking for recommendations · Comment on restaurants/dishes
- Open restaurant pages directly from posts
- Save restaurants to personal/shared collections · Revisit via saved lists

USER WORKFLOW

Browse community posts / friend recs → open linked restaurant pages → read dish comments & trust signals → save/share to lists → return later through collections.

Why it matters: Makes discovery more socially trusted, conversational, and engaging than review-only platforms.

Feature Workflow: Search → Filter → Compare Cards → Explore Community → Details

Key Interactive Elements

dynamic filtering & sorting · clickable restaurant cards → detail pages · list / map view toggle · community posts & discussion threads · save / favorite / shared lists

Key Features and Functionalities (2/2)

Rankings, Dish-Level Insights & Decision Support

CORE

Feature 3: Restaurant & Dish-Level Rankings

FUNCTIONALITY Helps users make better dining decisions through curated restaurant rankings and dish-level insights — answering not only "Which restaurant should I choose?" but also "What should I order there?"

INFO PROVIDED

Restaurant-Level Rankings

- Nearby Hot Picks · Top by Cuisine
- Community Favorites · Trending This Week
- Best for brunch, dinner, dessert, café, late-night

Dish-Level Insights (per restaurant)

- Top-rated dishes · Signature menu items
- User-uploaded dish photos
- Short reasons why a dish is recommended

INTERACTIVE

- Browse rankings by cuisine, area, or dining scenario
- Click ranking results → open restaurant detail pages
- Explore dish-level recommendations · View user-uploaded dish photos
- Upvote dishes or mark favorites · Save or share restaurant choices

USER WORKFLOW *Browse ranking by cuisine/area/scenario → open restaurant page → review top dishes, signature items & user photos → decide best restaurant + dish → save or share final choice.*

Why it matters: A major differentiator — supports both restaurant selection and dish-level decision-making in one platform.

SUPPORT

Feature 4: Menu Translation, Dish Explanation & Multilingual Menu Support

FUNCTIONALITY Helps users understand unfamiliar menu items across Asian cuisines through multilingual menu access, dish explanations, and translated food descriptions, lowering the barrier to trying new dishes.

INFO Translated menu names · Short dish explanations · Culturally specific food descriptions · Multilingual menu tabs (e.g. English, Simplified Chinese, Traditional Chinese, Japanese, Korean, Thai)

INTERACTIVE

- switch between multiple menu languages
- Open translation from restaurant detail page
- Tap/click unfamiliar dishes for explanation
- Use translation support while browsing menus

USER WORKFLOW *Users open a restaurant page → browse recommended or signature dishes → switch to their preferred language → read translated dish names and short descriptions → better understand menu options*

Why: Improves usability & accessibility in multilingual / cross-cultural dining situations.

Feature Workflow: Browse Rankings → Check Top Dishes → Validate Choice → Save / Share / Revisit

Key Interactive Elements

click rankings → detail pages · dish-level recommendation browsing · dish upvoting / favoriting · menu translation access · save / share restaurant choices

User Personas

Emily

22 years old

International Student from China in NYC

Brooklyn, NY

Background

Emily is a full-time college student juggling classes, part-time work, and social activities. She's always on the go and needs quick, affordable dining options that fit her budget and busy schedule.

Goals

- ✓ Find affordable restaurants near campus quickly
- ✓ Discover student-friendly deals and discounts
- ✓ Get quick meal suggestions between classes

Challenges

- ✗ Limited budget for dining out
- ✗ No time for extensive restaurant research
- ✗ Balancing social dining with saving money

Technical Familiarity

High - Native mobile user, uses apps daily

Preferred Devices

Mobile

Jason

29 years old

Asian-American Food Explorer in NYC

Manhattan, NY

Background

Jason is an Asian-American food enthusiast who loves exploring authentic Asian cuisine across NYC. He actively seeks out traditional dishes and trending fusion spots, always looking for the best ramen, dim sum, and Korean BBQ.

Goals

- ✓ Discover authentic Asian restaurants and hidden gems
- ✓ Explore diverse Asian cuisines (Chinese, Japanese, Korean, Thai, Vietnamese)
- ✓ Get recommendations based on specific dish preferences

Challenges

- ✗ Too many generic restaurant recommendations
- ✗ Difficulty finding truly authentic Asian cuisine
- ✗ Needs menu details and ingredient information

Technical Familiarity

Very High - Early adopter, comfortable with all features

Preferred Devices

Mobile, Desktop

Yuki

26 years old

Short-Term Visitor from Japan Exploring NYC

Visiting New York City

Background

Yuki is visiting New York for two weeks and wants to experience both authentic Asian food that reminds her of home and trendy Asian fusion restaurants unique to NYC. She relies heavily on her phone for navigation and recommendations.

Goals

- ✓ Find restaurants with familiar Asian flavors while traveling
- ✓ Discover Asian restaurants near tourist attractions
- ✓ Access menus in multiple languages or with photos

Challenges

- ✗ Unfamiliar with NYC neighborhoods and transit
- ✗ Language barriers at some restaurants
- ✗ Limited time to explore during short trip

Technical Familiarity

High - Comfortable with mobile apps and maps

Preferred Devices

Mobile

Technical Approach

FRONTEND

React + HTML / CSS / JavaScript

- **HTML** structures pages.
- **CSS** handles styling and responsive layout across devices.
- **React** builds reusable components: search bar, filter panel, restaurant cards, ranking list, map view, community interaction modules.
- **JavaScript** manages client-side interaction logic and enables dynamic updates without page reloads.

MAP INTEGRATION

Leaflet.js + OpenStreetMap

- Use Leaflet.js with OpenStreetMap to implement a lightweight, open-source mapping solution that supports marker-based restaurant visualization, user geolocation, and dynamic map updates driven by frontend filtering logic.

DATABASE

MongoDB

- Schema-less model — ideal for varied attributes, nested reviews & photos.
- Stores restaurants, reviews, users & favorites; 2dsphere index for location queries.

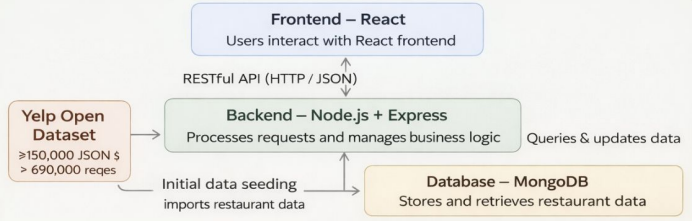
BACKEND

Node.js + Express Server

Central layer connecting React frontend to MongoDB

- Node.js serves as the server-side runtime, while Express handles routing and API management.
- **Data Seeding (Stage 1)** — a one-time Node.js script parses Yelp JSON files and batch imports restaurant and review data into MongoDB.
- **API Server (Stage 2)** — express provides RESTful APIs for frontend communication via HTTP, exchanging data in JSON format and supporting core features such as restaurant search and filtering, paginated reviews, user authentication, and favorites management.

Users interact with the React interface, which sends HTTP requests to Express API endpoints. The backend processes search, filtering, review, and favorites logic, while MongoDB stores and retrieves restaurant and review data. Results are returned in JSON format for dynamic rendering on the frontend.



Team Roles & Responsibilities

Asian Restaurant Discovery Platform · 5 Members · Responsibilities evenly distributed across Frontend, Map, Backend & Database

MEMBER 1

FRONTEND LEAD

[Jialing Wang]

React · Component Architecture · State Management

- Build the core React app structure: routing, folder architecture, and shared component library
- Implement the search bar, filter panel, and structured restaurant card system
- Manage application-wide state and ensure smooth, responsive user interactions

MEMBER 2

FRONTEND UI

[Yan Lu]

React · Dynamic Filtering · Sorting · Real-time Updates

- Own Figma wireframes and interaction refinements; ensure UI aligns with the prototype
- Implement dynamic sorting, real-time filtering, and favorites interfaces
- Build the community feed and dish-level detail views; support UI polish and accessibility testing

MEMBER 3

MAP & LOCATION EXPERIENCE

[Yu-Chien Chen]

Leaflet.js · OpenStreetMap · Responsive Design

- Integrate Leaflet.js + OpenStreetMap: restaurant markers, user geolocation, and cluster views
- Build list/map view toggle and ensure smooth transitions between display modes
- Ensure responsive design and cross-device compatibility across all screens and breakpoints

MEMBER 4

BACKEND DEVELOPMENT

[Meirui Yu]

Node.js · Express · REST APIs · Authentication

- Design and implement RESTful API endpoints for all core features: search, filter, reviews, auth, favorites
- Handle business logic: pagination, query parameter parsing, session management, and error handling
- Connect frontend requests to DB operations; ensure data consistency and secure route protection

MEMBER 5

DATABASE & DATA ENGINEERING

[Victoria Liu]

MongoDB · Yelp Dataset · Geospatial Indexing · Query Optimization

- Design MongoDB schemas for restaurants, users, reviews, and favorites; define indexes and relationships
- Build data pipeline: parse and batch-import Yelp JSON dataset (150K+ restaurants, 690K+ reviews)
- Optimize query performance: 2dsphere geospatial indexing, compound indexes, and filtering efficiency

ALL MEMBERS *Shared:*

- End-to-end testing & bug fixing
- Figma prototyping, UX flow refinement, and design-system alignment
- Presentation preparation & demo walkthrough

Timeline & Milestones

PROJECT TIMELINE

Mar 31 · NOW

Proposal Submission

All slides finalized & submitted · Figma wireframes v1 complete

Apr 5

Data Seeding & Backend Setup (Stage 1)

GitHub monorepo (/client, /server) · MongoDB setup & DB connection · Node.js Yelp JSON parsing & import to MongoDB · 2dsphere index · Express setup

Apr 10

API Server & Frontend Components (Stage 2)

Server-side API development: Express RESTful APIs for search, filter, paginated reviews, user authentication, and favorites · Client-side development: React components · Leaflet.js map scaffolded

Apr 18

Full Integration & Map Live

Frontend ↔ Backend fully connected · Leaflet.js map live with restaurant markers & geolocation · Reviews, ratings & favorites done

Apr 23

Testing & Polish

End-to-end testing + bug fixes · Responsive design QA + UI polish · Final deployment

Apr 29 · DEADLINE

Final Presentation

Conclusion & Next Steps

KEY TAKEAWAYS

🎯 Clear Purpose

An **asian-focused** and **community-powered** restaurant discovery platform for international students & food explorers **in NYC and nearby NJ**

⚙️ Solid Tech Stack

React + Node.js + Express + MongoDB (2dsphere) + Leaflet.js — fully self-hosted, geospatially optimized

👥 User-Centered

Three personas—a Chinese international student, an Asian-American food explorer, and a Japanese short-term visitor—represent our core users and anchor every design and feature decision support

💡 Main Functionalities

Smart filtering, Community interaction, Dish-level rankings, and Multilingual menu support

IMMEDIATE NEXT STEPS

01

Finalize Figma Wireframes

Complete all page prototypes and align with team on design system

02

GitHub Repo Setup

Initialize monorepo with /client and /server, configure branches

03

MongoDB Data Seeding

Collection structure (restaurants, reviews, users) → Yelp JSON parsing → data validation → 2dsphere index setup

04

Distribute Apr 5 Dev Tasks

Assign API endpoints and React components across the team